

Assignment No. 8

Subject: Database Engineering Lab (7CS271)

Class: SY CSE, AY 2025–26, Sem-II

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Batch - S5

Views and Authorization Mechanisms

Q. Implement Views and Authorization Mechanisms in DBMS . A relational database has been created for the selected case study system. The database contains multiple tables such as users, transactions, operational records, and administrative data.

Different types of users require restricted and controlled access to the database .To achieve this:

- Views are used to present customized data.
- Authorization mechanisms are used to control access

Task 1: Create Simple Views

a. Display limited columns from a table

Query

```
CREATE VIEW Student_Basic AS  
  
SELECT student_id, semester, cgpa  
  
FROM Student;
```

Result

STUDENT_ID	SEMESTER	CGPA
103	6	7.8
101	4	8.5
102	4	9
104	6	8.2
105	8	7.5

b. Hide sensitive information (e.g., salary, payment details)

Query

```
CREATE VIEW Person_Public AS  
  
SELECT person_id, first_name, last_name, city  
  
FROM Person;
```

Result

PERSON_ID	FIRST_NAME	LAST_NAME	CITY
6	Ankit	Verma	Pune
7	Neha	Joshi	Mumbai
8	Rahul	Kapoor	Delhi
9	Priya	Nair	Chennai
10	Vikram	Reddy	Hyderabad
3	Rohan	Mehta	Mumbai
5	Karan	Singh	Delhi
1	Aarav	Sharma	Pune
2	Diya	Patel	Pune
4	Sneha	Iyer	Bangalore

Task 2: Create Complex Views Create views involving:

a. Multiple tables using JOIN

Query

```
CREATE VIEW Student_Course_View AS

SELECT s.student_id, p.first_name, c.title

FROM Student s

JOIN Person p ON s.person_id = p.person_id

JOIN Enrolls e ON s.student_id = e.student_id

JOIN Course c ON e.course_id = c.course_id;
```

Result

STUDENT_ID	FIRST_NAME	TITLE
103	Rohan	DSA
105	Karan	SE
101	Aarav	DBMS
102	Diya	OS
104	Sneha	CN

b. Aggregation functions (SUM, AVG, COUNT)

Query

```
CREATE VIEW Course_Enrollment_Count AS
SELECT course_id, COUNT(*) AS total_students
FROM Enrolls
GROUP BY course_id;
```

Result

COURSE_ID	TOTAL_STUDENTS
504	1
502	1
505	1
503	1
501	1

Task 3: Updatable Views

a. Create a view that allows INSERT/UPDATE operations

Query

```
CREATE VIEW Student_Update_View AS  
  
SELECT student_id, cgpa  
  
FROM Student;
```

b. Perform update operations through the view

Query

```
UPDATE Student_Update_View  
  
SET cgpa = 9.5  
  
WHERE student_id = 101;
```

Result

```
1 row(s) updated.  
  
0.01 seconds
```

c. Verify changes in base table

Query

```
SELECT * FROM Student WHERE student_id = 101;  
  
And  
  
SELECT * FROM Student_Update_View;
```

Result

STUDENT_ID	PERSON_ID	ENROLLMENT_DATE	GRADUATION_DATE	SEMESTER	CGPA
101	1	07/01/2022	-	4	9.5

STUDENT_ID	CGPA
103	7.8
101	9.5
102	9
104	8.2
105	7.5

Task 4: Views with Conditions

a. Create a view using WHERE clause

Query

```
CREATE VIEW High_CGPA_Students AS
```

```
SELECT * FROM Student
```

```
WHERE cgpa > 8;
```

Result

STUDENT_ID	PERSON_ID	ENROLLMENT_DATE	GRADUATION_DATE	SEMESTER	CGPA
101	1	07/01/2022	-	4	9.5
102	2	07/01/2022	-	4	9
104	4	07/01/2021	-	6	8.2

b. Restrict rows based on conditions (e.g., active users, high-value transactions) Authorization Mechanisms

Query

```
CREATE VIEW Active_Students AS
```

```
SELECT * FROM Student
```

```
WHERE graduation_date IS NULL;
```

Result

STUDENT_ID	PERSON_ID	ENROLLMENT_DATE	GRADUATION_DATE	SEMESTER	CGPA
103	3	07/01/2021	-	6	7.8
101	1	07/01/2022	-	4	9.5
102	2	07/01/2022	-	4	9
104	4	07/01/2021	-	6	8.2
105	5	07/01/2020	-	8	7.5

Task 5: Grant Privileges

Use GRANT command to:

- a. Allow SELECT access to a user on specific tables

Query

```
GRANT SELECT ON Student TO user1;
```

Note: We can't run this command due to privilege restrictions in Oracle APEX environment.

OR WE CAN DO THIS

```
GRANT SELECT ON Student TO PUBLIC;
```

Result

```
Statement processed.
```

```
0.03 seconds
```

- b. Allow INSERT and UPDATE on selected tables

Query

```
GRANT INSERT, UPDATE ON Student TO PUBLIC;
```

Result

```
Statement processed.
```

```
0.01 seconds
```


c. Grant access only to specific columns

Query

```
GRANT SELECT ON Student TO PUBLIC;
```

Result

```
Statement processed.
```

```
0.14 seconds
```

Task 6: Revoke Privileges

Use REVOKE command to:

a. Remove SELECT permission

Query

```
REVOKE SELECT ON Student FROM PUBLIC;
```

Result

```
Statement processed.
```

```
0.02 seconds
```

b. Remove UPDATE/INSERT permissions

Query

```
REVOKE INSERT, UPDATE ON Student FROM PUBLIC;
```

Result

Statement processed.

0.05 seconds

c. Verify restricted access

Query

```
SELECT * FROM Student;
```

Result

STUDENT_ID	PERSON_ID	ENROLLMENT_DATE	GRADUATION_DATE	SEMESTER	CGPA
103	3	07/01/2021	-	6	7.8
101	1	07/01/2022	-	4	9.5
102	2	07/01/2022	-	4	9
104	4	07/01/2021	-	6	8.2
105	5	07/01/2020	-	8	7.5

5 rows returned in 0.02 seconds

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Task 7: Role-Based Access Control

Create roles such as: ADMIN MANAGER USER

Tasks:

- a. Assign appropriate privileges to each role

Query

```
CREATE ROLE ADMIN;
```

```
CREATE ROLE MANAGER;
```

```
CREATE ROLE USER_ROLE;
```

- b. Assign roles to users

Query

```
GRANT ALL PRIVILEGES TO ADMIN;
```

```
GRANT SELECT, INSERT, UPDATE ON Student TO MANAGER;
```

```
GRANT SELECT ON Student TO USER_ROLE;
```

- c. Verify access restrictions for each role

Query

```
GRANT ADMIN TO PUBLIC;
```

```
GRANT MANAGER TO PUBLIC;
```

```
GRANT USER_ROLE TO PUBLIC;
```

“CREATE ROLE requires DBA privileges which are not available in APEX environment.”